GRIDINTEL · UTILITY INTELLIGENCE

Reliability and service standards

SAIDI, SAIFI and CAIDI by feeder and substation, attribution by where interruptions began, service-band compliance by day, and the DisCo's reported figures against the calculation on their basis.

SAMPLE: SYNTHETIC DATA

This report is a sample. It shows the format and the methods GridIntel would deliver; it is not a report of real results. Every figure in it is calculated from a designed demonstration dataset, and none describes a real network, site, customer or organization.

A designed demonstration dataset for September 2026. It describes no real network, customer, meter reading, outage, bill or payment.

FOR
Regulator; the DisCo's network planning

RECORDS OF
Savanna Electricity Distribution (synthetic)

PERIOD
14 September 2026

REPORT ID
GI-REL-2026-09-14-C45001B9

GENERATED
15 Oct 2026, 14:06 WAT

DATA AS OF
15 Sep 2026, 00:00 WAT

DEFINITION
Version 1

DATA FINGERPRINT (SHA-256)
75a86faba855c32ac47d8b592d988da652dafae3f9d16416e22ac41c924b331d

IN BRIEF

SAIDI
8.1 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

SAIFI
1.5 • C
OK · Calculated · gridintel.reliability.reference 0.3.0

CAIDI
5.4 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

SAIDI ATTRIBUTED TO THE DISTRIBUTION NETWORK
0.0 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

CUSTOMERS SERVED
6,324 • C
OK · Calculated

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1 The DisCo's indices

How often, and for how long, the average customer was interrupted, and how much of it was the distribution network's.

SAIDI
8.1 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

SAIFI
1.5 • C
OK · Calculated · gridintel.reliability.reference 0.3.0

CAIDI
5.4 h • C
OK · Calculated · gridintel.reliability.reference 0.3.0

ASAI
66.3% • C
OK · Calculated · gridintel.reliability.reference 0.3.0

CUSTOMERS SERVED
6,324 • C
OK · Calculated

By attribution: whose interruption it was

Each interruption is attributed by where it began, under the reference methodology. The rows add up to the totals above.

	SAIDI (H)	SAIFI	SHARE OF CUSTOMER-HOURS
Network Faults, planned work and weather on the distribution network, including the lines that feed its substations.	0.0 h • C	0.0 • C	0.0% • C
Upstream supply Loss of supply that began at the transmission station or on the grid.	0.0 h • C	0.0 • C	0.0% • C
Load shedding Supply withheld under load management.	8.1 h • C	1.5 • C	100.0% • C
Other Customer, third party, or not known.	0.0 h • C	0.0 • C	0.0% • C

Method: GridIntel reference reliability indices (gridintel.reliability.reference 0.3.0).

2 By substation

The same indices for each injection substation and the customers it serves.

Indices by substation

All interruptions, then the part attributed to the distribution network.

	CUSTOMERS	SAIDI (H)	SAIFI	CAIDI (H)	NETWORK SAIDI (H)	NETWORK SAIFI
Riverside 33/11 kV injection substation	3,543 • C	6.5 h • C	1.6 • C	4.1 h • C	0.0 h • C	0.0 • C
Hillcrest 33/11 kV injection substation	2,781 • C	10.1 h • C	1.4 • C	7.4 h • C	0.0 h • C	0.0 • C

Method: GridIntel reference reliability indices (gridintel.reliability.reference 0.3.0).

3 By feeder

Which feeders fail their customers, ranked by what the distribution network itself did.

Indices by 11 kV feeder

Ranked by network-attributable SAIDI, highest first. Loss of upstream supply and load shedding are shown beside it and do not set the rank.

	CUSTOMERS	SAIDI (H)	SAIFI	CAIDI (H)	NETWORK SAIDI (H)	UPSTREAM SAIDI (H)	LOAD SHEDDING SAIDI (H)
1. Farm Road 11 kV feeder Hillcrest 33/11 kV injection substation	1,015 • C	12.0 h • C	2.0 • C	6.0 h • C	0.0 h • C	0.0 h • C	12.0 h • C
2. Government Avenue 11 kV feeder Hillcrest 33/11 kV injection substation	1,766 • C	9.0 h • C	1.0 • C	9.0 h • C	0.0 h • C	0.0 h • C	9.0 h • C
3. Market Road 11 kV feeder Riverside 33/11 kV injection substation	1,477 • C	3.0 h • C	1.0 • C	3.0 h • C	0.0 h • C	0.0 h • C	3.0 h • C
4. Old Town 11 kV feeder Riverside 33/11 kV injection substation	2,066 • C	9.0 h • C	2.0 • C	4.5 h • C	0.0 h • C	0.0 h • C	9.0 h • C

Method: GridIntel reference reliability indices (gridintel.reliability.reference 0.3.0).

4 Where interruptions began

Attribution by the part of the system each interruption began at, which decides whether it is the distribution network's to fix.

By origin point

The part of the system the outage record names as where the interruption began, and which side of the boundary it is on.

	SAIDI (H)	SAIFI	SHARE OF CUSTOMER-HOURS
The grid (supply allocated) Load management or upstream	8.1 h • C	1.5 • C	100.0% • C

Method: GridIntel reference reliability indices (gridintel.reliability.reference 0.3.0).

By cause

The cause the outage record gives.

	SAIDI (H)	SAIFI	SHARE OF CUSTOMER-HOURS
Load shedding	8.1 h • C	1.5 • C	100.0% • C

Method: GridIntel reference reliability indices (gridintel.reliability.reference 0.3.0).

The elements interruptions most often began at

Interruptions other than load shedding, most interruptions first.

> No interruption other than load shedding is recorded in the period.

Interruptions other than load shedding, by the element the outage record names as where they began. A 33 kV line and the transmission station are not registry assets, so they keep the names the outage log gives them.

5 Service-band compliance

Whether each feeder gave its customers the hours of supply its service band commits to, day by day.

Hours of supply against the band, by feeder

Each day is tested against the band minimum on its own; the average over the period is shown beside it and is a separate test.

	BAND	BAND MINIMUM	AVERAGE SUPPLIED	DAYS MET	DAYS BELOW
Farm Road 11 kV feeder Hillcrest 33/11 kV injection substation	Band D	8.0 h/day • C	12.0 h/day • C	1 • C	0 • C
Government Avenue 11 kV feeder Hillcrest 33/11 kV injection substation	Band B	16.0 h/day • C	15.0 h/day • C	0 • C	1 • C
Market Road 11 kV feeder Riverside 33/11 kV injection substation	Band A	20.0 h/day • C	21.0 h/day • C	1 • C	0 • C
Old Town 11 kV feeder Riverside 33/11 kV injection substation	Band C	12.0 h/day • C	15.0 h/day • C	1 • C	0 • C

Method: GridIntel reference hours of supply (gridintel.supply_hours.reference 0.2.0).

Farm Road 11 kV feeder: GridIntel reference calculation — not a regulatory figure. Band minimums are NERC Service-Based Tariff definitions.

Government Avenue 11 kV feeder: GridIntel reference calculation — not a regulatory figure. Band minimums are NERC Service-Based Tariff definitions.

Market Road 11 kV feeder: GridIntel reference calculation — not a regulatory figure. Band minimums are NERC Service-Based Tariff definitions.

Old Town 11 kV feeder: GridIntel reference calculation — not a regulatory figure. Band minimums are NERC Service-Based Tariff definitions.

Days below the band minimum

Every feeder-day that fell short, by feeder.

	DAY	HOURS SUPPLIED	BAND MINIMUM
Government Avenue 11 kV feeder Band B	2026-09-14	15.0 h • C	16.0 h/day • C

Method: GridIntel reference hours of supply (gridintel.supply_hours.reference 0.2.0).

6 Interruptions still open at the end of the period

Which figures are provisional because a restoration is not yet recorded.

> No interruption in the records was open at the end of the period. No figure in this report is provisional for that reason.

Feeders whose figures are provisional

An interruption still open is counted to the end of the period. The figure is calculated, not estimated, and is not final until the restoration is recorded.

> None.

7 Reported against calculated

Each figure the DisCo reported for this period, beside GridIntel's calculation on the reported figure's own basis, and what a report's attribution rule changes.

The DisCo's reported figures

> No reported figure in the records is for 14 September 2026.

> **Reports for other periods.** The records also hold reported figures for 1 September 2026 to 30 September 2026. They are not compared here: a reported figure is compared only with a calculation for the same period.

8 Method, sources and how to reproduce this report

Where every figure came from, and how a reader can check that this document is the one the records give.

Methodologies

METHODOLOGY	VERSION	WHAT IT SAYS OF ITSELF
GridIntel reference reliability indices `gridintel.reliability.reference`	`0.3.0`	GridIntel reference calculation — not a regulatory figure.
GridIntel reference hours of supply `gridintel.supply_hours.reference`	`0.2.0`	GridIntel reference calculation — not a regulatory figure. Band minimums are NERC Service-Based Tariff definitions.

Sources

- Synthetic outage log (synthetic)
- Synthetic asset registry (synthetic)

> Every source above is synthetic: designed demonstration data, not records of a real organization.

How to reproduce it

1. Build the report with definition version 1, for 14 September 2026, from the Savanna Electricity Distribution (synthetic) records.
2. The data fingerprint must be 75a86faba855c32ac47d8b592d988da652dafae3f9d16416e22ac41c924b331d. SHA-256 over the sorted SHA-256 digests of every distinct result the report's services were given by the repositories, each written as JSON with its keys in order.
3. With the same definition, methodologies and fingerprint the report ID is the same: GI-REL-2026-09-14-C45001B9. A different ID means something above differs.
4. The figures with their status, origin, unit and method are in the CSV of the same report ID: gridintel-reliability-2026-09-14-SAMPLE-SYNTHETIC.csv.

9 Regulatory references

> How far this report carries the fields of the instruments below, as read from the instruments themselves. It is not a claim that the report conforms to any regulator's or programme's template: none has been checked against the template it would be filed on.

INSTRUMENT	REQUIREMENT	THIS REPORT
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	1. SAIDI: total duration of interruption (hrs), total number of interruptions to customers, total number of registered customers; monthly	**PARTLY** Sections 1–3 give SAIDI and its parts by DisCo, substation and feeder. The denominator is customers served by the network in the registry; whether it is the Schedule's "registered customers" is to be confirmed.
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	2. SAIFI: total number of system interruptions, total number of supply points affected (from distribution transformer uprisers); monthly	**PARTLY** Sections 1–3 give SAIFI per customer served. A count by supply point (transformer upriser) is not calculated.
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	3. CAIDI: SAIDI, SAIFI; monthly	**CARRIED** Sections 1–3.
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	4. CAIFI: total number of customer interruptions, total number of registered customers; monthly	**NOT CARRIED** Not calculated: it needs the number of distinct customers interrupted, which the methodology does not yet count.
NERC Reporting Compliance Regulations 2009 (NERC-R-02-09), s. 9 and Schedule 4A	5–6. HV and LV fault clearance indices: faults cleared within 8 hours against faults reported; monthly	**NOT CARRIED** Not calculated. The outage records hold the times it would be counted from.
NERC June 2024 Supplementary Order to MYTO 2024 (e.g. ORDER/NERC/2024/068, YEDC), para. 9 and Appendix 1	Service-Based Tariff bands and their minimum hours of supply a day (A 20, B 16, C 12, D 8, E 4)	**CARRIED** Section 5 tests each feeder's days against its band minimum, and gives its average over the period.

INSTRUMENT	REQUIREMENT	THIS REPORT
NERC June 2024 Supplementary Order to MYTO 2024 (e.g. ORDER/NERC/2024/068, YEDC), para. 9 and Appendix 1	9(c). Publish daily a rolling seven-day average of daily hours of supply on each Band A feeder	**PARTLY** Section 5 and the CSV give each Band A feeder's hours of supply day by day; the rolling seven-day average is not calculated.
NERC June 2024 Supplementary Order to MYTO 2024 (e.g. ORDER/NERC/2024/068, YEDC), para. 9 and Appendix 1	9(d)–(e). A Band A feeder below its committed level on two, and on seven, consecutive days	**PARTLY** Section 5 lists every feeder-day below the band minimum; runs of consecutive days are not counted.

10 Glossary and key

SAIDI	System Average Interruption Duration Index: the hours of interruption the average customer served had in the period. Customer-hours interrupted ÷ customers served.
SAIFI	System Average Interruption Frequency Index: the interruptions the average customer served had in the period. Customer interruptions ÷ customers served.
CAIDI	Customer Average Interruption Duration Index: the average length of an interruption, in hours. SAIDI ÷ SAIFI.
ASAI	Average Service Availability Index: the share of customer-hours in the period with supply.
Network-attributable	The part of an index caused by interruptions that began on the distribution network (33 kV and 11 kV lines, distribution transformers, the low-voltage network), under the reference methodology.
Upstream supply	Interruptions that began at the transmission station or above it: not the distribution network's.
Load shedding	A feeder opened on purpose because the supply allocated was short. It began nowhere on the network.
Origin point	The part of the system an outage record names as where an interruption began. It decides the attribution.
Service band	NERC's Service-Based Tariff band of a feeder, with its minimum average hours of supply a day: A 20, B 16, C 12, D 8, E 4.
Provisional	A figure that includes an interruption still open at the end of the period, counted to its end. It is calculated, not estimated, and is final only when the restoration is recorded.
Attribution-rule finding	Where a reported figure states an attribution rule other than the reference one: the same figure on both rules and the difference, so a reader sees what the rule changes.
Measured	Read from a meter, a device or a source record.
Reported	A figure published by someone else, shown as they stated it. It is never altered.
Calculated	Computed by GridIntel from records, under a named and versioned methodology.
Estimated	An estimate, not a measurement; the method is named with it.
Derived	A residual: what is left when other figures are subtracted. It carries their uncertainty.
Insufficient data	An input the figure needs is missing. No value is given rather than a guess, and never a zero.
Not available	No source holds this figure, or the data does not cover the period.
Same basis	A reported figure and a calculation count the same things over the same period by the same rule. Only figures on the same basis are compared.
Data fingerprint	A SHA-256 digest of every record the report was built from. The same records give the same fingerprint; any change to them gives another.
Report ID	The report's code, its period and the first characters of a digest of its definition, methodologies and data fingerprint. The same report of the same data has the same ID.

The marks beside a figure

• **OK**	Every input is present and measured.
≈ **Estimated inputs**	Calculated; every input is present, but at least one was estimated rather than measured.
! **Insufficient data**	An input this figure needs is missing. No value is shown rather than a guess.
∅ **Not computable**	The inputs exist but the calculation is undefined for them.
– **Not available**	The platform has no source for this figure.
M **Measured**	Read from a meter, device or source record.

R **Reported**	Published by someone else; shown as stated.
C **Calculated**	Computed by GridIntel under a named methodology.
E **Estimated**	An estimate, not a measurement.
D **Derived**	A residual: what is left after subtracting other figures. It inherits their uncertainty.

SAMPLE: SYNTHETIC DATA. End of report GI-REL-2026-09-14-C45001B9.